

SIMATS ENGINEERING

Assessment - CO4-AT3

Subject: Artificial Intelligence

Course Code: CSA 1712

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Course Outcome Covered (CO4) — Article Review

Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Task 1: Article Summary

(a) Article Citation, Domain & ML Problem Addressed

Bhat, S. S., Banu, M., Ansari, G. A., & Selvam, V. (2023). A risk assessment and prediction framework for diabetes mellitus using machine learning algorithms. *Healthcare Analytics*, 4, Article 100273, Elsevier.

DOI / Link: <https://doi.org/10.1016/j.health.2023.100273>

Domain / Application Area: Healthcare / clinical decision support — early risk assessment and diagnosis of diabetes mellitus (covering Type 1, Type 2, and Gestational forms) from routine patient clinical measurements.

Specific ML Problem Addressed: Binary supervised classification — predicting whether a patient is diabetic or non-diabetic from clinical features, benchmarked across three distinct machine learning algorithms to identify the most reliable model for risk prediction.

(b) Summary of the Article (Own Words)

This study addresses the growing global burden of diabetes mellitus by building and comparing three machine learning classifiers — Logistic Regression, Gradient Boosting, and Decision Tree — to predict diabetes risk from routine clinical measurements. Using the widely available Pima Indian Diabetes dataset of 768 patient records, the authors evaluate each model with multiple statistical performance metrics rather than relying on accuracy alone, and find that the Decision Tree classifier gives the strongest overall performance. The research gap the authors identify is that much existing diabetes-prediction literature evaluates only a single algorithm or a narrow set of metrics, and rarely frames the problem holistically across the different clinical forms of diabetes (Type 1, Type 2, and Gestational), making it hard to judge which modelling approach is genuinely most trustworthy for real screening use. The authors fill this gap by systematically benchmarking three distinct algorithm families — a linear statistical model, an ensemble boosting method, and a tree-based inductive learner — on the same dataset and metric suite, producing a comparative risk-assessment framework intended to guide model selection for practical diabetes screening tools.

Task 2: Methodology Analysis

(a) ML Techniques Used & Dataset Description

Logistic Regression: a statistical learning model that estimates the log-odds of a patient being diabetic as a linear combination of input features, producing an interpretable probability output.

Gradient Boosting: an ensemble technique that builds a sequence of weak learners (typically shallow decision trees), each one correcting the errors of the previous ensemble, to progressively minimise prediction loss.

Decision Tree: a tree-structured inductive classifier that recursively splits the feature space on the attribute that best separates diabetic from non-diabetic patients (e.g. using Gini Index/Information Gain), producing an interpretable set of if-then rules — this was the best-performing model in the study.

Dataset: the Pima Indian Diabetes Dataset — 768 patient records sourced from the UCI (University of California, Irvine) Machine Learning Repository, each with clinical attributes (e.g. glucose concentration, blood pressure, BMI, insulin level, age, number of pregnancies, skin-fold thickness, diabetes pedigree function) and a binary outcome label (diabetic / non-diabetic). The authors prepared the data by applying standard preprocessing and

then evaluated the three algorithms using multiple statistical performance metrics on the resulting train/test split.

(b) Mapping to CO4 Topics, Strength & Limitation

Mapping to CO4: the Decision Tree component maps directly to the course's Decision Tree Algorithms topic; Logistic Regression maps to Statistical Learning Methods; and the overall process of inducing a general classification rule from labelled patient examples (across all three models) reflects the course's Inductive Learning topic, with the end goal matching CO4's focus on building models for Classification and Decision-Making.

Methodological Strength: benchmarking three structurally different algorithm families (linear statistical, ensemble boosting, and tree-based) on the same dataset with multiple statistical metrics gives more trustworthy, well-rounded evidence for model selection than a typical single-model study, reducing the risk of over-crediting one algorithm due to metric or dataset quirks.

Methodological Limitation: the study relies on a single, small, publicly available benchmark dataset (768 records, drawn only from female patients of Pima heritage), which limits how confidently the reported performance generalises to broader, more demographically diverse patient populations encountered in real clinical deployment.

Task 3: Results & Critical Evaluation

(a) Key Results

As reported in the article's own highlighted findings, the Decision Tree classifier was the best-performing model:

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree (best model)	91%	96%	92%	94%
Gradient Boosting	Evaluated; underperformed Decision Tree	—	—	—
Logistic Regression	Evaluated; underperformed Decision Tree	—	—	—

Note: the article's abstract and highlights explicitly report the Decision Tree's accuracy, precision, recall, and F1-score (91%, 96%, 92%, and 94% respectively) as the best result across the three models; the exact per-metric figures for Logistic Regression and Gradient Boosting individually were not detailed in the accessible portions of the article, so they are reported here only as 'underperforming' relative to the Decision Tree, in the interest of not misrepresenting figures the authors did not explicitly publish in the accessible summary.

(b) Critical Evaluation

Realism of reported metrics: a 91% accuracy on the 768-record Pima Indian Diabetes dataset is plausible and broadly consistent with the wider literature using the same benchmark, where comparable studies report

Decision Tree/ensemble accuracies in a similar 80–95% range — so the headline figures are credible rather than implausibly inflated.

Potential Dataset Bias: the Pima dataset contains only female patients of Pima Native American heritage, so any model trained on it risks poor generalisation to male patients or other ethnic populations; the dataset is also well known in the literature for containing biologically implausible zero values (e.g. zero blood pressure or BMI) that represent missing data, and if these were not carefully treated, they could distort the reported metrics.

Potential Evaluation-Protocol Bias: if performance was assessed on a single train/test split rather than repeated k-fold cross-validation, the reported metrics could carry optimistic variance from a favourable split rather than reflecting the model's true average performance.

Additional Experiments to Strengthen Findings: external validation on an independent, more demographically diverse cohort; k-fold cross-validation with confidence intervals rather than a single-split score; explicit class-imbalance handling (e.g. SMOTE) with before/after comparison; explainability analysis (e.g. SHAP or LIME) to justify which features drive the Decision Tree's splits; and a comparison against stronger ensembles such as Random Forest or XGBoost to confirm the Decision Tree's advantage holds against more competitive baselines.

(c) Real-World Applicability

Deployment Settings: a lightweight, interpretable Decision Tree-based screening tool of this kind could be deployed in primary-care clinics, community health screening camps, or integrated into electronic health record (EHR) systems as a decision-support prompt flagging patients for confirmatory diabetes testing, and is well suited to low-resource settings given its low computational footprint and rule-based interpretability that clinicians can directly inspect.

Data Availability: scaling beyond a single 768-record benchmark requires access to larger, more representative, and ethically sourced patient datasets spanning diverse populations — a significant practical hurdle given healthcare data privacy restrictions.

Compute & Integration: while training these particular models is computationally light, real-world deployment requires integration with hospital EHR pipelines, data-quality pipelines, and ongoing model monitoring infrastructure, which is a bigger engineering undertaking than the standalone research prototype.

Ethical Issues: patient data privacy and consent (e.g. HIPAA/GDPR-equivalent compliance), risk of biased predictions for populations under-represented in the training data, and the need for clear accountability if an automated screening flag is missed or wrongly raised, are all necessary considerations before clinical-grade deployment.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights (Connected to CO4 Concepts)

Insight 1 — Tree-based Inductive Learning Captures Non-linear Clinical Thresholds: seeing the Decision Tree outperform Logistic Regression on this tabular clinical data deepened my understanding of why tree-based inductive learning (CO4: Decision Tree Algorithms / Inductive Learning) can capture non-linear threshold effects — e.g. a glucose cut-off — that a purely linear statistical model struggles to represent directly.

Insight 2 — Multiple Metrics Matter More Than Accuracy Alone in High-Stakes Classification: the article's use of accuracy, precision, recall, and F1-score together (rather than accuracy alone) reinforced the course's emphasis on evaluating classification models properly (CO4: Decision Trees / Classification), especially since a high accuracy can still hide a poor recall — clinically dangerous for a disease-screening task.

Insight 3 — Statistical and Tree-based Learning Are Complementary, Not Competing, Tools: benchmarking Logistic Regression against Decision Tree and Gradient Boosting side-by-side illustrated that CO4's Statistical Learning Methods and Decision Trees topics are not rival techniques to pick once and for all, but complementary tools whose relative strength (interpretability vs. non-linear fit) should be evaluated empirically for each specific dataset and clinical use case.

(b) Proposed Extension: Novel Experiment / Enhancement

Proposed Enhancement: extend this comparative Logistic Regression / Gradient Boosting / Decision Tree framework to a larger, more demographically diverse Gestational Diabetes Mellitus (GDM) dataset covering multiple ethnicities and both sexes where applicable, paired with a Reinforcement Learning-based longitudinal lifestyle-intervention recommender (echoing CO4's Reinforcement Learning topic) trained on repeated follow-up visits rather than a single diagnostic snapshot, and adding SHAP-based explainability to the winning classifier.

Feasibility: publicly available, more diverse gestational and general diabetes datasets exist on repositories such as Kaggle and the UCI Machine Learning Repository, and open-source SHAP/LIME libraries make explainability analysis straightforward to add to an existing scikit-learn pipeline, so this extension is realistically achievable with modest additional engineering effort over the original study's methodology.

Expected Impact: moving beyond a single, demographically narrow snapshot dataset to a more diverse and longitudinal one would directly address the generalisability limitation identified in the critical evaluation, while adding a Reinforcement Learning component would shift the system's value from a one-off diagnostic flag to an ongoing, adaptive treatment-support tool — and SHAP-based explainability would make the resulting model's recommendations more trustworthy and clinically actionable for physicians.